Questions by topic: The circular economy, biocapacity & ecological footprints - Set 1 (50 marks)

Answer all questions in the spaces provided.

0 1 . 1 Describe the main principles of the circular economy.

[6 marks]

0 1 . 2 Explain what is meant by 'optimum production rather than maximum production', using a comparison between a natural ecosystem and an agroecosystem.

[2 marks]

0 1 . 3 Explain why international cooperation (eg through the EU or UN) is needed to implement circular economy initiatives.

[2 marks]

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0 2 . 1 Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

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Extra space

0 2 . 2 State what is meant by **industrial symbiosis**.

[1 mark]

10

0 3 Table 1 shows the ecological footprint and biocapacity per person for four countries and the world in 2016.

Table 1

	Ecological footprint / gha per person	Biocapacity / gha per person	Human Development Index
USA	8.1	3.6	0.92
UK	4.4	1.4	0.92
China	3.6	1.0	0.75
Bangladesh	0.8	0.4	0.61
World	2.8	1.7	0.72

0 3 . 1 Define **ecological footprint** and **biocapacity**.

[2 marks]

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0 3 . 2 Explain what is meant by a **global hectare** and why it is used.

[2 marks]

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0 3 . 3 Use Table 1 to calculate how many Earths would be needed if everyone lived like the average US citizen.

[1 mark]

Answer _____ Earths

0 3 . 4 Use Table 1 to describe the relationship between development and ecological footprint, and explain the concept of Earth Overshoot Day.

[3 marks]

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0 3 . 5 Suggest **two** ways a country such as the UK could reduce its footprint without reducing its HDI.

[2 marks]

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0 4 . 1 Explain what a carbon footprint should include when calculated for a car.

[3 marks]

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0 4 . 2 Describe what the WWF Living Planet Index measures and how it is used.

[2 marks]

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0 4 . 3 Describe how the Ol Pejeta reserve in Kenya combines commercial cattle ranching with wildlife conservation.

[5 marks]

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0 5 The Isle of Eigg (area 31 km^2, population about 100) is not connected to the national grid. Table 1 shows its electricity sources.

Table 1

Source	Capacity / kW
Hydroelectric (3 generators)	110
Solar PV	50
Wind (4 turbines)	24
Diesel generators (back-up)	160

0 5 . 1 Explain why Eigg uses a mix of three renewable resources rather than one, and why diesel back-up is still needed.

[3 marks]

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0 5 . 2 Describe **three** other ways in which the Eigg community has increased its sustainability.

[3 marks]

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0 5 . 3 Explain why small self-contained communities often find it easier to adopt sustainable lifestyles than large cities.

[2 marks]

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0 5 . 4 Explain how large-scale DC electricity grids such as Desertec could increase the use of renewable energy.

[2 marks]

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END OF QUESTIONS